

Review 1

(1) Is there a market for the proposed book? Is the book both topical and timely, either for use at undergraduate, graduate or research level, or as a reference?

The proposed has a very specific target market in mind: undergraduate students of ecology. It is topical in the sense that it tackles a classical subject in many biology-related undergraduate courses: ecology. Its contributions are thus timeless and the focus on providing the added R guidance make it very relevant to contemporary ecology students in dire need of quantitative training.

How long do you predict that the book will remain usefully up to date?

As mentioned in my previous comment, the subject covered is timeless and the different concepts covered have been the subject of ecological textbooks for decades. As those previous works, this book should remain useful and up to date for a few generations of ecologists. The fact that R is nowadays such an extensively used statistical and data manipulation computer language also suggests that will remain useful for a long time.

That being said, modern AI tools based on LLMs are now making possible to auto generate code to do many tasks. This has created some degree of laziness in students who don't want to learn these tools and they just ask ChatGPT to give them an answer (i.e. write code for them). This might jeopardize the relevance of learning coding skills in years to come.

Who, in your opinion, would use this book?

Undergraduate students desiring to learn quantitative methods for ecology from a basic entry level. Instructors wanting to expose their students to these analytical tools and exploration of relevant ecological content from an applied perspective, using tangible examples that can make the learning more effective.

Please provide further details of the courses for which this book could be used, if appropriate, and outline how relevant the book would be for such a course? How widespread are these courses?

This book could be used for 1st year ecology courses. The book would constitute a relevant complement to (to be used alongside) classical ecology textbooks. It achieves this by providing real-world examples of ecological studies and the accompanying R code illustrating how to tackle ecological questions quantitatively and develop the associated computer code to analyse the corresponding data.

Courses like this are widespread as introductory ecology courses form the basis of the curriculum of virtually all ecology focused degrees worldwide.

2) Are the contents, organization, and level of the book appropriate to the intended audience? Is the author attempting to serve too many different readers? If so, on which audience should they focus?

I believe the content, organization and level of the book to be appropriate to the intended audience. The author is attempting to serve an appropriate audience (1st year ecology students), making it a very focused and specific text made for a very specific and clear purpose.

Does the author(s) cover the subject fully, engaging with all essential sides of the topic and with work from a diverse range of perspectives? Is there any obvious bias?

The main argument and the methodology sections of the proposal were very short and lack any specific detail, so it was hard to evaluate this part fully.

Related to this, the “Audience” paragraph is an almost verbatim copy of the “Main argument” section. This could be remedied by having a more comprehensive argument section offering a better understanding of the topics of the book and how they come together to generate an integral ecology book.

Just based on the limited information presented in the document, and the proposed TOC it seems like the subject is fully covered and the essential aspects of ecology (individual behaviour, the physical environment, demographics, density dependence and ecological interactions) are covered.

Is the planned content representative of the field in all other aspects of its diversity? If not, how could it be improved?

The planned content is representative of what is expected to be covered in an introductory ecology course for undergraduates.

One aspect that could be improved in chapter 12 is that the concepts and theories related to spatial ecology, which are mentioned only in reference to what was learned in the lecture, can be better presented and explained in more detail. Concepts of dispersal and connectivity, and in the context of metapopulations and metacommunities could be elaborated on.

Does the book incorporate sufficient geographic diversity to ensure international appeal?

I cannot comment on this as I am not sure how to evaluate geographic diversity. That being said, ecology being a universal science, the concepts learned here are of applicability to virtually any ecosystems around the world, and include the main tools ecologists use to understand ecosystems across biomes.

(3) Is the author well-placed to achieve their intended purpose?

Based on the previous experience of the author, his international research profile, and the fact that he has authored a couple of books in the discipline before, I consider the author to be well placed to deliver the proposed book and its content.

(4) Are there any other titles either existing or in preparation which compete directly with this book?

I believe that the list of related titles provided by the author is a fair representation of the field in terms of the books that are currently out there. Insofar as these titles represent a partial

overlap with the current proposed book, the perspective provided by the book in terms of very applied and tangible ecological issues explained at an entry level, accompanied by the corresponding computer code to illustrate the desk-based research task and how to learn it, will, I believe, ensure that it competes only minimally with existing titles.

(5) Do you recommend the publication of this book? If so, is your recommendation conditional on any particular modifications?

Yes, I would recommend publication of this book.

I have offered a suggestion for improvement in the “how could it be improved?” question, but will leave to the discretion of the editor whether this should condition publication.

Would you recommend it to your library?

Yes, I would recommend it to my library.

Would you recommend it to your students?

Yes, I would recommend it to my first year undergraduate students.

Would you buy a copy yourself? (a hypothetical question since you will receive a complimentary copy on publication as a token of thanks for this review).

I personally won't need a copy of a book like this, but I would buy it for teaching purposes.

(6) Have you any other comments which may help to improve this proposal?

Review 2

(1) Is there a market for the proposed book? Is the book both topical and timely, either for use at undergraduate, graduate or research level, or as a reference?

This is a very interesting and novel ecology textbook which uses an enquiry-based approach to teach ecological concepts. Few if any ecology textbooks use a problem-based approach, which is mostly the domain of data analysis textbooks. I think there is definite market for this, especially to run more novel ecology curricula that center on skill acquisition. It is an appropriate accompaniment for an undergraduate ecology course and could be easily incorporated into practical courses. The proposed ready-made activities would be attractive for busy ecology instructors looking to match their content with practical skills sessions.

How long do you predict that the book will remain usefully up to date?

The case studies in the book span the last 20 years or so. Some chapters are tied to existing published datasets in data repositories which are stable. Others rely on dynamic datasets existing in apps and may need risk-assessment for permanence. Climate projections will need data updates minimum once a decade to remain relevant. This could be done in a digital-only format by updating datasets only, while the printed book is worded carefully around it.

Who, in your opinion, would use this book?

I believe this book would be popular with ecology course instructors at undergraduate level. This is very much a workbook and could be used to run a skills based part of an ecology course.

Please provide further details of the courses for which this book could be used, if appropriate, and outline how relevant the book would be for such a course? How widespread are these courses?

This book could be used in undergraduate ecology courses which are very common at UK universities, as well as in European and African institutions where English is the didactic language. I have no experience of the US HE landscape, but given the pick-and-mix nature of their undergraduate degrees, I suspect ecology would be a very common STEM subject at an undergraduate level.

2) Are the contents, organization, and level of the book appropriate to the intended audience? Is the author attempting to serve too many different readers? If so, on which audience should they focus?

This book is aimed at undergraduate learners and maintains its focus throughout. Its chapter titles are definitely geared towards attracting attention from students, as though the author is teaching an elective course. I had to stop and think a while to map some of the chapter titles onto ecological principles.

Does the author(s) cover the subject fully, engaging with all essential sides of the topic and with work from a diverse range of perspectives? Is there any obvious bias?

The coverage is thorough and covers topical issues. There is quite a lot of process modelling as opposed to pattern analysis in the activities, which is less common in practical undergraduate guides which tend to focus on statistics.

Is the planned content representative of the field in all other aspects of its diversity? If not, how could it be improved?

There is a fairly good spread from behavioural ecology to ecosystems ecology. Perhaps one thing that could be added is a representation of microbial ecology beyond bacterial symbionts of the pea plant.

Does the book incorporate sufficient geographic diversity to ensure international appeal?

There is good geographical spread, including datasets from tropical ecosystems in S America, Asia, and Africa. There are also global issues such as human population growth and climate projections, which I think are going to be appealing.

(3) Is the author well-placed to achieve their intended purpose?

The author has clearly tried out this content over the course of his teaching practice as a higher education professional and is well-placed to write this book.

(4) Are there any other titles either existing or in preparation which compete directly with this book?

Not to my knowledge.

(5) Do you recommend the publication of this book? If so, is your recommendation conditional on any particular modifications?

Yes, I think this book could be a very useful asset to ecology instructors globally. Its focus on open source data and tools is particularly important to reach global audiences. The only risk is the risk of obsolescence of some of the app data used in chapter one, which needs to be assessed.

Would you recommend it to your library?

Yes

Would you recommend it to your students?

Yes

Would you buy a copy yourself? (a hypothetical question since you will receive a complimentary copy on publication as a token of thanks for this review).

Yes

(6) Have you any other comments which may help to improve this proposal?

One of the chapters features a photo of the researcher, which I think could be used throughout the book to showcase the human element and the diversity of people working in ecology. This could be inspirational for students.

Review 3

(1) Is there a market for the proposed book? Is the book both topical and timely, either for use at undergraduate, graduate or research level, or as a reference?

Yes, there is a huge need for R competency to be developed early on in ecology. There is a steep learning curve for anyone interested in learning R, and students of current and upcoming generations entering undergraduate institutions seem to have less ability to problem solve and trouble shoot technology. Students who hit a stumbling block give up very quickly.

Quantitative skills are weak in undergraduates and also often weak in faculty. There is a major demand for clear quantitative activities that can be readily incorporated into existing curriculum. This book could be applicable in undergraduate and graduate courses or to help students get going on their own independent use of R for research.

How long do you predict that the book will remain usefully up to date?

I have some reservations about any book that uses technology heavily to remain up to date in the long term (>10 years). The first chapter is an essential piece of the story (intro to your computer and R). While R changes very little in terms of its general function, computers can often go through dramatic shifts between different versions. Having worked with R for years, I've experienced issues finding working directories on new devices, and the use of one drive or other cloud based services is also a potential concern for users into the future. I'm not sure how digital books stay up to date, but I assume a new edition could be released with minor updates as needed and also include any updated links for the data locations and resources throughout. For example, many of the current links connect to Dr. Steven's google drive or access data through existing webpages. How long will those exist? Who knows.

Still, R is a technology that is not going away, and as a free, open source resource, users of R have lots of forums and resources at their disposal. Googling issues can readily get answers. Aside from The R pieces, the overall scientific thought process is clearly laid out in most of the chapters following a clear flow from hypothesis→conclusions. This provides students with examples of how ecological questions were answered. This is not something that goes out of date.

Who, in your opinion, would use this book?

The content of this proposed book could be used by undergraduates, graduates, or faculty who want to grow their skills in R and be able to apply them to ecological scenarios.

Please provide further details of the courses for which this book could be used, if appropriate, and outline how relevant the book would be for such a course? How widespread are these courses?

This course could be used in part or as a whole for undergraduate or graduate ecology courses, environmental science (possibly even for non-majors if only pieces are used). If the chapters could be more of standalone pieces, I could see that the first chapter if combined with one or two other pieces could fit into many courses within the biology curriculum at an undergraduate institution: Intro bio (especially the last part of the course that gets to more global/community level issues), Biostatistics, conservation biology, Environmental science, zoology. These courses are offered across the board at most undergraduate institutions with biology majors. I also could see it being used by undergraduate students who want to learn how to use R in preparation for graduate school or for ecological analyses but aren't sure where to start or have little guidance from their mentors.

2) Are the contents, organization, and level of the book appropriate to the intended audience? Is the author attempting to serve too many different readers? If so, on which audience should they focus?

The content and flow are similar for most of the chapters following scientific thought processes (steps of scientific method) and clearly laying out key goals for the chapter. The content spans multiple organizational levels of ecology, as an ecology course would, and the pieces would fit in (though maybe not everyone teaches the stuff in the same order). I think it is important that Dr. Steven's final version of this book be something that could be used in pieces for broader reach. If each chapter builds on the other (as explained in the preface), it may make it hard to adopt fully and fit into existing curriculum as easily. Maybe it would be possible for Dr. Stevens to connect or link a specific chapter/R code to prior key chapter pieces if necessary for understanding and moving forward. It may also be valuable to include a "teacher version" that provides key pitfalls and examples of what the correct answers are, so the faculty can verify answers easily and be prepared for potential issues that might arise. I would also suggest an idea of how long these activities might take- could they be completed during a lab period, Or would it be something that the students would work on outside of class?

Does the author(s) cover the subject fully, engaging with all essential sides of the topic and with work from a diverse range of perspectives? Is there any obvious bias?

I think the author covers a broad range of topics, vertebrates and invertebrates and to some extent geographic variation.

The authors begin the book with a piece about environmental justice, without having it directly relate to race or any potential controversial issue but simply related to income. I think this is a good approach. The work in this draft demonstrates the current scientific perspective of the facts in the field of ecology.

Is the planned content representative of the field in all other aspects of its diversity? If not, how could it be improved?

I think the book covers important topics and covers a range of organisms that sufficiently for the field of ecology. I mention below some suggestions related to geographic representation.

Does the book incorporate sufficient geographic diversity to ensure international appeal?

I think it is good in this regard. One potential area that it could improve is to maybe use data that is related to Asian or African continents. There is one chapter that covers work in Panama, but I think the majority of the data and studies focus on North America. A good connection for Asian and African pieces could fit into the population growth chapters or related to global emissions in the carbon cycle chapter.

(3) Is the author well-placed to achieve their intended purpose?

Yes- Dr. Stevens has spent much of his career communicating R to scientists, non-scientists, experts and non-experts. He has already practiced using these activities in courses and I think has honed them over the years.

(4) Are there any other titles either existing or in preparation which compete directly with this book?

I don't know of any. This is a well-known need in the field, with relatively few equipped to provide it.

(5) Do you recommend the publication of this book? If so, is your recommendation conditional on any particular modifications?

Yes- I think it would be great. No conditions and only a few recommendations, already explained above.

Would you recommend it to your library?

I think it would be valuable to have and I would recommend my library carry it.

Would you recommend it to your students?

I think I would recommend it by actually using it in the classroom. I might recommend it to a couple of my independent research students who are interested in research in ecology.

Would you buy a copy yourself? (a hypothetical question since you will receive a complimentary copy on publication as a token of thanks for this review).

I probably wouldn't buy a copy myself. I don't teach Ecology currently- this is where I would see the greatest application, but I think I would recommend it to a couple of faculty in my department to use in Ecology courses and also a biometrics course (biostatistics)

(6) Have you any other comments which may help to improve this proposal?

Dr. Stevens mentions that the book isn't really about statistics or about how to do statistical methods, which I agree, it's not really focused on that, but the ability to interpret statistical figures and outputs is important. Perhaps a short chapter or appendix that includes an overview of common graphs and statistics that will be found throughout would be a valuable addition.

I also would highly recommend a resource guide that would help faculty with facilitating each of the chapters if needed including any background knowledge they must be prepared with.

Review 4

(1) Is there a market for the proposed book? Is the book both topical and timely, either for use at undergraduate, graduate or research level, or as a reference?

Yes, there is a clear market for this book. An important feature is that it can supplement lessons from a larger, traditional textbook or serve as a workbook alongside an existing curriculum. The approach here is consistent with pedagogical research suggesting that active learning (i.e., learning by doing) is effective for engagement, learning, and retention. The book is both topical and timely as I believe many instructors would like to incorporate more active learning (especially, through guided data analysis using statistical software), and this book will provide ready to use activities through which to do it.

How long do you predict that the book will remain usefully up to date?

I believe the book will remain relevant for at least five years. At some point, changes to R packages used may require updating the guided activities somewhat, although this probably depends on which specific packages are incorporated. Likewise, there may be an eventual need to update the datasets/examples used to keep the guided examples somewhat current. In general, the structure of the book would not need to change and this could be accomplished via updated versions of the book. The need for guided, data driven examples and the centrality of the R language to ecology research are unlikely to change in coming years, so the book should have some longevity in that regard.

Who, in your opinion, would use this book?

Instructors/and students in undergraduate ecology classes are the primary audience for this, although it might even be useful for graduate courses. This should provide a large market for potential users.

Please provide further details of the courses for which this book could be used, if appropriate, and outline how relevant the book would be for such a course? How widespread are these courses?

This book will be useful primarily for general ecology courses, as the subject matter generally follows themes introduced in general ecology textbooks. In some cases, I could envision that instructors of more discipline specific courses (e.g., population ecology, community ecology) could use the book as a supplement. These are very widespread courses that are part of the core curriculum for students in a variety of fields that require application of ecology-based principles (e.g., biology, environmental biology, environmental science, conservation biology, fisheries/wildlife management, environmental economics, and many others)

2) Are the contents, organization, and level of the book appropriate to the intended audience? Is the author attempting to serve too many different readers? If so, on which audience should they focus?

The contents are appropriate and not too broad.

Does the author(s) cover the subject fully, engaging with all essential sides of the topic and with work from a diverse range of perspectives? Is there any obvious bias?

There is no obvious bias, and the author covers many core concepts in general ecology courses.

Is the planned content representative of the field in all other aspects of its diversity? If not, how could it be improved?

I do not think the book covers the full breadth of topics that could be covered, but given the broad nature of ecology, I think the choice of topics is appropriate and provides vignettes that are appropriate. For example, the latter part of the book (Chapters 9-12) is focused more on community ecology, but does not cover the full breadth of species interactions that are typically discussed in community ecology units, instead focusing on specific applications of community ecology principles (invasion potential through competition, mutualisms through an ant/plant interaction, and predator). This is not necessarily a problem if the book is used as a supplement to another text, but I do think it limits the book's utility as the sole text in a general ecology course. In general, the book is more focused on applications of ecological principles through vignettes.

I do not think it is a stand in for a more general textbook given the relative lack of broad background in the history of the field, general discussions of ecological principles, broad conceptual diagrams, etc., but this is not the author's aim; it would be an extremely valuable workbook to encourage active learning using specific examples, implemented using quantitative thinking in a coding language that is invaluable to students.

Does the book incorporate sufficient geographic diversity to ensure international appeal?

The book is North America centric. I do not think that this limits the utility of the case studies included, which are useful regardless of where students reside.

(3) Is the author well-placed to achieve their intended purpose?

Yes, Dr. Stevens is exceptionally well-placed to implement this, as evidenced by his quantitative ecology background, but especially by his experience authoring a previous, similar book aimed at more advanced students.

(4) Are there any other titles either existing or in preparation which compete directly with this book?

The author did a good job of summarizing similar texts in his proposal; I am not aware of others.

(5) Do you recommend the publication of this book? If so, is your recommendation conditional on any particular modifications?

I do recommend that this book is published, and do not have any particular critiques that must be addressed prior to consideration.

Would you recommend it to your library?

Yes.

Would you recommend it to your students?

Yes.

Would you buy a copy yourself? (a hypothetical question since you will receive a complimentary copy on publication as a token of thanks for this review).

I would buy a copy.

(6) Have you any other comments which may help to improve this proposal?

I wish the author luck with this idea and am grateful for their efforts to create ready-to-use, active learning activities that involve quantitative reasoning implemented in an important coding language. I believe instructors and students alike will benefit from the availability of this resource.